London, UK @ a.gallaghersyed@gmail.com Google Scholar GitHub amayags.github.io LinkedIn

Amaya **GALLAGHER-SYED**

ML Research Scientist | Bio x AI

SUMMARY

Machine learning researcher and computational biologist working at the intersection of AI and biology. I build explainable, biology-aligned models and probe what foundation models encode about biological structure, across single-cell genomics and computational pathology in immunology and cancer. My current focus is identifiable representation learning and cross-modal causal discovery, working toward mechanistic virtual-cell models.

RESEARCH EXPERIENCE

ERIC AND WENDY SCHMIDT AI FOR SCIENCE FELLOW — **2026.09–present**
Imperial College London
◇ Identifiable representation learning for biology foundation models, with applications in dynamical systems and cross-modal causal discovery.

POSTDOCTORAL RESEARCH FELLOW — **2025.11–2026.09**
Queen Mary University of London · Imperial College London
◇ Mechanistic probing of single-cell foundation models via differentiable gene regulatory network recovery from CRISPR Perturb-seq data.
◇ Explainable deep learning for cancer survival prediction.

WELLCOME TRUST DOCTORAL RESEARCHER — **2021.11–2025.09**
Queen Mary University of London
◇ Developed explainable, multimodal deep learning for computational pathology in autoimmune disease and cancer, spanning graph neural networks, attention-based multiple-instance learning, and image–omics fusion.

TEACHING ASSISTANT — **2021–2023**
Queen Mary University of London
◇ Statistics for AI and Data Science · Post-Genomics Bioinformatics.

TEACHING ASSISTANT — **2018–2019**
Universidad de Buenos Aires
◇ Botany & Plant Physiology.

EDUCATION

WELLCOME TRUST PHD IN SCIENCE — **2020–2025**
Queen Mary University of London
◇ Thesis: *Explainable Deep Learning for Multi-Stain and Multimodal Computational Pathology in Autoimmune Diseases*. Supervisors: M. R. Barnes · M. J. Lewis · G. Slabaugh.

MSC IN MATHEMATICAL SCIENCES — **2019–2020**
Queen Mary University of London
◇ Distinction. Thesis: *Machine learning for gene expression data of early Rheumatoid Arthritis patients to distinguish treatment responders from non-responders*.

LICENTIATE IN COMPUTATIONAL BIOLOGY — **2012–2019**
Universidad de Buenos Aires
◇ 8.3/10 (Honours). 7-year full-time degree spanning biology, chemistry, physics, mathematics, statistics, and programming.

SELECTED PUBLICATIONS

◇ **A. Gallagher-Syed**, A. Wenteler, S. A. Martinez, G. Slabaugh. **CLAMP: A Mechanistic Probe of Regulatory Structure in Foundation Models under Single-Cell Perturbations**, *ICML 2026 Workshop on Generative and Agentic AI for Biology. Spotlight (2026)*.
◇ **A. Gallagher-Syed**, M. J. Lewis, M. R. Barnes, G. Slabaugh. **ProtoPathway: Biologically Structured Prototype-Pathway Fusion for Multimodal Cancer Survival Prediction**, *Under Review* (2026).

- A. Wenteler, M. Occhetta, N. Branson, M. Huebner, V. Curean, W. Dee, W. Connell, A. Hawkins-Hooker, P. Chung, Y. Ektefaie, C. M. Cordova, **A. Gallagher-Syed**. **PertEval-scFM: Benchmarking Single-Cell Foundation Models for Perturbation Effect Prediction**, *International Conference on Machine Learning (ICML)* (2025).
- **A. Gallagher-Syed**, H. Senior, O. Alwazzan, E. Pontarini, M. Bombardieri, C. Pitzalis, M. J. Lewis, M. R. Barnes, L. Rossi, G. Slabaugh. **BioX-CPath: Biologically-driven Explainable Diagnostics for Multistain IHC Computational Pathology**, *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2025).
- J. Duquesne, L. Basseto, C. Claye, M. Barnes, E. Pontarini, **A. Gallagher-Syed**, et al. **Machine learning to classify the focus score and Sjogren's disease using digitalised salivary gland biopsies: a retrospective cohort study**, *The Lancet Rheumatology*, 7(12), e864-e872 (2025).
- **A. Gallagher-Syed**, E. Pontarini, M. J. Lewis, M. R. Barnes and G. Slabaugh. **Going Beyond H&E and Oncology: How Do Histopathology Foundation Models Perform for Multi-stain IHC and Immunology?**, *NeurIPS 2024 Workshop on Advancements In Medical Foundation Models* (2024).
- **A. Gallagher-Syed**, L. Rossi, F. Rivellese, C. Pitzalis, M. J. Lewis, M. R. Barnes, G. Slabaugh. **Multi-stain self-attention graph multiple instance learning pipeline for histopathology Whole Slide Images**, *34th British Machine Vision Conference (BMVC)* (2023).

FUNDING & AWARDS

- Eric and Wendy Schmidt AI for Science Postdoctoral Fellowship (≈£120k).
- Wellcome Trust Early Career Transition Fund (≈£40k).
- Wellcome Trust PhD Studentship in Science (≈£200k).
- Outstanding Reviewer, CVPR 2026.
- Top Reviewer (top 4%), NeurIPS 2024.

INVITED TALKS

- **Keynote — LatinX in AI Workshop @ ICML** **2025.07**
  From Buenos Aires to London: how broad scientific foundations shaped my AI research.
- **AI x Bio Conference — Wellcome Connecting Science** **2025.06**
  BioX-CPath: Biologically-driven Explainable Diagnostics for Multistain Computational Pathology.

TECHNICAL SKILLS

Machine learning: representation learning, foundation-model probing & benchmarking, transformers, graph/hypergraph neural networks, multimodal fusion, computer vision.
Modeling: differentiable programming (implicit differentiation), neural ODEs & dynamical systems, causal inference from interventional data, gene regulatory network inference.
Programming & infrastructure: Python (PyTorch, PyTorch Geometric/DGL), Scanpy/AnnData, Git, multi-GPU / distributed training (SLURM, HPC).
Biology: single-cell & spatial transcriptomics, CRISPR Perturb-seq, immunology, systems biology, computational pathology.
Languages: English, French, Spanish (all native).

ACADEMIC SERVICE

Reviewer: NeurIPS 2026, ECCV 2026, CVPR 2026, ICLR 2026, TMLR, CVPR 2025, ICLR 2025, NeurIPS 2024, MICCAI 2023.
Open source: contributor to the Graph Structure Learning Benchmark (NeurIPS 2023).
Organiser: DERI Lunch & Learn Seminar. **Co-organiser:** Responsible AI Workshop.